Day - 05

Calculating Gradients by Hand

Mathmatical Assignment:

Problem 1: Symbolic partial Derivatives.

$$f(x,y) = 3x^2y + \sin(x)e^y$$

Task A: Calc the partial derivatives of x, denoted a $\frac{\partial f}{\partial x}$

Task B: " " " " y " $\frac{\partial f}{\partial y}$

Solution:

Task A:

$$\frac{\partial f}{\partial x} = 3.2xy + \cos(x).e^y = 6xy + e^y\cos(x)$$

Task B:

$$\frac{\partial f}{\partial y} = 3x^2 + e^y\sin(x)$$

Problem 2: evaluation the gradient vector.

Task A: evaluate $\frac{df}{dx}$ at $x=2, y=1$

Task B: " $\frac{\partial f}{dy}$ at $x=2, y=1$

Task C: Construct gradient vector $\nabla f(2,1)$.

Solution:

CamScanner

Task A: $\frac{\partial f}{\partial x}(2,1) = 6.2.1 + e^1 \cos(2)$

$= 12 + e\cos(2)$

Task B: $\frac{\partial f}{\partial y}(2,1) = 3.2^2 + e^1 \sin(2)$

$= 12 + e\sin(2)$

Task C: $\nabla f(2,1) = \begin{bmatrix} 12 + e\cos(2) \\ 12 + e\sin(2) \end{bmatrix}$

<u>Problem 3:</u> The Directional Derivatives.

you decide a unit vector $u = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$

Task A: Find the eqn for the derivatives $D_u f(2,1)$ by taking dot product of <u>gradient vector</u> $\nabla f(2,1)$. and <u>unit vector</u>.

<u>Solution:</u>

$D_u f(2,1) = \nabla f(2,1) . u$

$= \begin{bmatrix} 12 + e\cos 2 \\ 12 + e\sin \end{bmatrix} \cdot \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$

$= 16.8 + e(0.6\cos(2)) + 0.8\sin(2)$